Sheena Selvakumar

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sheena-2002.github.io/portfoli0 | *Silicon Validation / Analog IC Engineer Intern*

Education

Aug 2025 – May 2027	**M.S. Electrical Engineering**, *Texas A&M University*, GPA: 3.3/4.0. Coursework: Advanced Analog IC Design, CMOS RF Circuits, VLSI Design, Computer Architecture, Hardware Verification.
Jun 2020 – Jun 2024	**B.Tech. Electronics & Communication Engineering**, *PSG College of Technology*, India. GPA: 3.47/4.0

Work Experience

Jan 2026 – Present **Research Assistant — Silicon Photonics**, *Texas A&M University* (Advisor: Prof. Kareem Fayaz)

- Design and characterize integrated photonic devices; develop and execute bench validation plans using Python-automated data extraction pipelines (Lumerical FDTD/COMSOL) to generate structured test reports and drive design iteration — directly analogous to silicon bench validation workflows.
- Apply device-physics intuition to interpret simulation results, identify failure modes, and propose corrections; workflow mirrors silicon bring-up, electrical characterization, and sign-off validation methodology used in ASIC physical design.

Jul 2024 – Jun 2025 **Analyst**, *Deloitte USI*, Chennai, India

- Collaborated with cross-functional teams (Design, Test, program management) to analyze system-level requirements, perform structured debugging, and deliver validated production solutions; led test execution across multiple environments applying root-cause analysis to improve reliability.
- Developed detailed technical documentation and engineering reports, presented findings to stakeholders, and contributed to continuous improvement of validation methodologies.

Project Work

Analog IC Design **Fully Differential Op-Amp, IBM 180nm CMOS** *Aug–Dec 2025*

- Designed telescopic op-amp with integrated CMFB; achieved >60 dB DC gain, 120 MHz GBW, ≥60° phase margin; performed electrical characterization and performance benchmarking across PVT corners via Monte Carlo mismatch analysis and post-layout PEX.

RF / Mixed-Signal **Inductor-less Quadrature Receiver, 2.4 GHz (ECEN 665)** *Jan 2026–Present*

- Implemented full 2.4 GHz RX chain (LNTA + passive mixer + TIA); met all specs: >40 dB gain, <3 dB NF, IIP3 > −5 dBm, IIP2 >30 dBm, <20 mW power; applied gm/ID methodology on IBM 180nm.

Data Converters **SAR ADC Design, 250 MHz 8 ENOB (ECEN 607)** *Jan 2026–Present*

- Designed SAR ADC (comparator, bootstrapped switches, cap DAC) at transistor level; 250 MHz BW, 8 ENOB, FoM >170 dB; performance benchmarking verified post-layout across corners in Cadence Spectre.

RF IC Design **CMOS LNA Design for Zigbee RF Transceiver** *Jan–May 2026*

- Designed and simulated a 1.9 GHz CMOS cascode LNA for Zigbee front-end in Cadence Spectre using realistic on-chip inductors; characterized I/O matching networks for optimal power transfer to on-chip and off-chip loads (50Ω).

HW Verification **Functional Verification of HTAX (HyperTransport Crossbar)** *Aug–Dec 2025*

- Developed UVM-based automated test framework in SystemVerilog/Cadence Xcelium; constrained-random validation scripts identified a latent RTL bug; achieved **100% functional coverage** and ≥95% block/expression coverage via vManager regressions.

Computer Arch. **L3 Cache & Memory Controller Policy Evaluation (PACIPV & EAF)** *Aug–Dec 2025*

- Benchmarked DDR-interfacing L3 cache/memory controller policies vs. LRU/SRRIP using ZSim (C++) on PARSEC/SPEC CPU2006 (8-core, 16 MB LLC); built Python automation scripts to run simulations, parse logs, and generate IPC/MPKI performance benchmarking reports.

Technical Skills

Programming	Python (test automation, NumPy/Matplotlib), C/C++, MATLAB, SystemVerilog, Verilog, Linux Shell
Lab & Validation	Oscilloscopes, multimeters, power supplies, AWGs, BERT; bench validation planning; electrical characterization; PVT corner & Monte Carlo analysis
EDA Tools	Cadence Virtuoso ADE, Spectre (DRC, LVS, PEX), Xcelium, vManager, SPICE, Xilinx Vivado, Lumerical FDTD, COMSOL, HFSS
Analog / RF	Fully Differential Op-Amps, CMFB, RF Front-Ends, SAR ADC, LDO/Regulator Fundamentals, DDR/Memory Interface awareness, Common-Centroid Layout, Frequency Compensation

Professional Development

Mixed-Signal IC	Full design cycle experience spanning specification, circuit design, simulation, post-layout verification (DRC/LVS/PEX), and silicon bring-up; bench validation plan development and execution; PVT corner validation; automated bench testing (Python/MATLAB); performance benchmarking and data-driven analysis; familiar with DDR/memory interface validation, EM/IR considerations, and translating silicon measurements into actionable design improvements.
Modeling & Verification	Developing proficiency in Verilog-A/AMS for behavioral modeling of ADCs, DACs, and PLLs to bridge circuit- and system-level mixed-signal evaluation workflows; designed silicon photonic devices (waveguides, directional couplers, resonators, power splitters) for high-speed optical communication using EM simulation tools.
Product Lifecycle	Motivated to contribute across the full mixed-signal IC product lifecycle — from specification definition and circuit design through silicon validation, characterization, and customer-facing application engineering.

Certifications

2026	Cadence Semiconductor 101 | 2025: SystemVerilog for Design and Verification | Digital Electronics